SOCIOL 723: Social Statistics II

Week 13, Causal Machine Learning

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Today

Why Naive ML Fails

Double Machine Learning

Heterogeneous Treatment Effects

★ = essential, you must understand this ★ = for understanding, not examined

Putting the Two Halves Together ★

- **Weeks 5–11** gave us identification: what must be true for a parameter to be causal. Almost every design ended with a nuisance function to estimate, a propensity score, an outcome regression, a first stage, a conditional trend.
- **Week 12** gave us flexible estimators for exactly such functions, which work with many covariates and unknown functional form.
- The obvious move, plug ML into the causal estimator, fails. Today is about why, and what fixes it.

The payoff

We get to relax the functional-form assumptions that identification arguments have been quietly leaning on all semester, AJR's linear geography control, the linear propensity score, the additive conditional trend, *without* giving up $\sqrt{n}$ inference.

The Problem

The Partially Linear Model ★

To see the problem without letting notation hide it, start with the simplest causal case there is: *one* treatment effect to estimate, but completely flexible relationships between the controls, the treatment, and the outcome. That is the partially linear model, the workhorse for the rest of the course:

$$\begin{aligned} Y_i &= \theta\, D_i + g(W_i) + \epsilon_i, \qquad & E[\epsilon_i \mid D_i, W_i] = 0 \\ D_i &= m(W_i) + v_i, \qquad & E[v_i \mid W_i] = 0. \end{aligned}$$

- θ is the target: the effect of D holding W fixed, and it is assumed constant (relaxed in the last section).
- g and m are nuisance functions; we do not care about them, but we must handle them. They may be highly nonlinear, and W may be high-dimensional.
- Under conditional ignorability given W (Week 5), θ is the ATE.
- Week 1 solved this when g and m are linear and low-dimensional. The question now is what happens when we estimate them with lasso or a forest.

Attempt 1: Regularization Bias ★

The naive approach: fit the partially linear model by ML to get $\hat{g}$, taking it from a joint fit of Y on (D_i, W_i) with D_i left *unpenalized* so that $\hat{g}$ targets the structural g. Then regress $Y_i - \hat{g}(W_i)$ on D_i.

$$\hat{\theta}_{\text{naive}} = \frac{\frac{1}{n}\sum_i D_i(Y_i - \hat{g}(W_i))}{\frac{1}{n}\sum_i D_i^2}.$$

Substituting the true model and rescaling,

$$\sqrt{n}(\hat{\theta}_{\text{naive}} - \theta) = \underbrace{\left(\frac{1}{n}\sum_i D_i^2\right)^{-1} \frac{1}{\sqrt{n}}\sum_i D_i\epsilon_i}_{\xrightarrow{d} \mathcal{N}(0,\Sigma)} + \underbrace{\left(\frac{1}{n}\sum_i D_i^2\right)^{-1} \frac{1}{\sqrt{n}}\sum_i D_i(g(W_i) - \hat{g}(W_i))}_{\text{regularization bias}}.$$

- Regularized estimators are *deliberately biased* (Week 12), and typically converge more slowly than the parametric $n^{-1/2}$.

Attempt 1: Regularization Bias ★ (cont.)

- Then the second term does not vanish on the $\sqrt{n}$ scale, the nuisance bias is of the same order as, or larger than, the sampling noise we are trying to describe.
- The point estimate may be fine; the inference is not. Confidence intervals are centred in the wrong place, and collecting more data does not fix it.
- **A trap worth naming.** If you instead regress Y on W alone you estimate $\ell(W) = E[Y \mid W] = \theta\, m(W) + g(W)$, which already contains the effect you are after. Subtracting that removes the signal. And even with the true ℓ, regressing $Y - \ell(W)$ on *raw* D_i does not recover θ: D_i must be residualized too, which is exactly what the next slide does.

Attempt 2: Partial Out Both Sides ★

Frisch–Waugh–Lovell says: residualize Y *and* D on W, then regress. With ML estimates $\hat{g}$ and $\hat{m}$,

$$\tilde{Y}_i = Y_i - \hat{\ell}(W_i), \qquad \tilde{D}_i = D_i - \hat{m}(W_i), \qquad \hat{\theta} = \frac{\frac{1}{n}\sum_i \tilde{D}_i \tilde{Y}_i}{\frac{1}{n}\sum_i \tilde{D}_i^2},$$

where $\ell(W) = E[Y \mid W]$.

The bias term now involves the product of the two estimation errors:

$$\text{bias} \approx \sqrt{n} \cdot \frac{1}{n} \sum_i \big(m(W_i) - \hat{m}(W_i)\big)\big(\ell(W_i) - \hat{\ell}(W_i)\big).$$

This vanishes when $\|\hat{m} - m\|_2 \, \|\hat{\ell} - \ell\|_2 = o_p(n^{-1/2})$, for which it is sufficient that *each* converge faster than $n^{-1/4}$. Lasso and forests can do that under sparsity or smoothness assumptions. **First-order bias has become second-order.**

This Is Neyman Orthogonality Again ★

The residualized moment condition is

$$E\Big[\psi(W_i;\theta,\eta)\Big] = 0, \qquad \psi = \Big[(Y-\ell(W)) - \theta(D-m(W))\Big]\,(D-m(W)),$$

and it satisfies

$$\frac{\partial}{\partial\eta} E\big[\psi(W_i;\theta_0,\eta)\big]\Big|_{\eta=\eta_0} = 0.$$

- You met this property in Week 6: the AIPW estimator's bias was the *product* of the two nuisance errors, so first-order mistakes in either one cancel. The partially linear case is the same statement for a different moment.
- And you derived it in **Week 1**: FWL is the linear special case, in which $\hat{m}$ is an OLS projection and the "adaptivity" of $\hat{\beta}_1$ to estimation error in the residuals is exactly orthogonality.
- **The thread of the whole course**: FWL $\to$ AIPW $\to$ DML. One idea, three levels of generality.

DML

The Second Ingredient: Cross-Fitting ★

Orthogonality is not quite enough. If $\hat{m}$ is fitted on the *same* observations where it is evaluated, its errors correlate with those units' outcomes, overfitting bias, which orthogonality does not remove.

Cross-fitting

Split the sample into K folds. For each fold k: estimate $\hat{\ell}^{(-k)}, \hat{m}^{(-k)}$ on the *other* folds, and form residuals only for observations *in* fold k. Then pool.

$$\hat{\theta}_{\text{DML}} = \frac{\frac{1}{n}\sum_k \sum_{i\in\mathcal{I}_k} \left(D_i - \hat{m}^{(-k)}(W_i)\right)\left(Y_i - \hat{\ell}^{(-k)}(W_i)\right)}{\frac{1}{n}\sum_k \sum_{i\in\mathcal{I}_k} \left(D_i - \hat{m}^{(-k)}(W_i)\right)^2}$$

This is the same device as cross-validation in Week 12, but used to protect *inference*, not to select a tuning parameter.

The DML Recipe ★

1. Split the sample into K folds (typically $K = 5$; split on the *independent unit* if data are clustered).
2. On the training folds, estimate $\hat{\ell}(W) = \hat{E}[Y \mid W]$ and $\hat{m}(W) = \hat{E}[D \mid W]$ with any ML method, lasso, random forest, boosting, or an ensemble.
3. On the held-out fold, form residuals $\tilde{Y}_i$ and $\tilde{D}_i$.
4. Pool across folds and regress $\tilde{Y}$ on $\tilde{D}$.
5. With independent observations, the standard error is the usual heteroskedasticity-robust one from that final bivariate regression: orthogonality is what makes it valid despite the ML first stage. *With clustered data, keeping clusters intact within folds is not enough*, the score variance must also be clustered at the independent unit.

The DML Recipe ★ (cont.)

6. Repeat the whole split several times with different seeds and take the median; the answer should not depend on the partition.

Chernozhukov et al. (2018). In `R`: `DoubleML` with `mlr3` learners.

What the Theory Requires, and What It Does Not ★

Requires

- Conditional ignorability given W (unchanged!).
- Overlap: $\hat{m}(W)$ bounded away from 0 and 1 for binary D.
- Nuisance convergence fast enough that the *product* of the two rates is $o(n^{-1/2})$; each faster than $n^{-1/4}$ is sufficient but not necessary. Approximate sparsity for lasso, smoothness for forests.
- Orthogonal moment + cross-fitting.

Does not require

- Correct functional form for g or m.
- Knowing which of the p covariates matter.
- $p \ll n$. High-dimensional W is the point.
- Consistent *model selection*, lasso may pick the wrong variables and $\hat{\theta}$ is still fine.

What the Theory Requires, and What It Does Not ★ (cont.)

The assumption that does not go away

DML does not identify anything. It is an estimation technology for a parameter already identified by the arguments of Weeks 5–11. If W omits a confounder, no amount of machine learning helps.

Where DML Plugs In

The same two ingredients, an orthogonal score plus cross-fitting, apply to almost every design in this course:

Design	Nuisance functions	Week
Partially linear regression	$E[Y \mid W]$, $E[D \mid W]$	1
Interactive / ATE under CIA (AIPW)	$E[Y \mid D, W]$, $\Pr(D = 1 \mid W)$	6
Instrumental variables	$E[Y \mid W]$, $E[D \mid W]$, $E[Z \mid W]$	7
Regression discontinuity	$E[Y \mid W]$ within the bandwidth	9
Conditional DID	outcome trend and propensity, by cohort	10

- For IV: residualize Y, D, *and* Z on W using ML, then run 2SLS on the residuals. This is what lets us replace AJR's "linear in distance from the equator plus continent dummies" with something flexible.

Where DML Plugs In (cont.)

- For conditional DID: this is precisely the doubly robust Callaway–Sant'Anna estimator from Week 10, with ML nuisances.

HTE

From ATE to CATE ★

Drop the constant-effect assumption. The target becomes the conditional average treatment effect

$$\tau(x) = E\big[Y_i(1) - Y_i(0) \mid X_i = x\big].$$

- Sociologically this is often the substantive question: *for whom* does college pay off, *which* neighbourhoods benefit, *who* is harmed.
- The naive approach, run subgroup regressions, report the significant ones, is a multiple-testing disaster and does not replicate. It is the specification search of Week 2 with more rooms to search.
- The modern approach: build an orthogonal, doubly robust *pseudo-outcome* whose conditional mean is $\tau(x)$, then use ML to estimate that conditional mean, with honest sample splitting so that the subgroups are not chosen using the same data that tests them.

The Doubly Robust Score as a Pseudo-Outcome ★

Take the AIPW summand from Week 6, unit by unit:

$$\Gamma_i = \hat{m}_1(X_i) - \hat{m}_0(X_i) + \frac{D_i(Y_i - \hat{m}_1(X_i))}{\hat{e}(X_i)} - \frac{(1 - D_i)(Y_i - \hat{m}_0(X_i))}{1 - \hat{e}(X_i)}.$$

- Its *average* is the AIPW estimate of the ATE; that was Week 6.
- With the *true* nuisance functions, its conditional mean is exactly the CATE: $E[\Gamma_i \mid X_i = x] = \tau(x)$. With cross-fitted estimates it holds approximately, at a rate governed by the same product condition as before.
- So: construct Γ_i with cross-fitted nuisances, then regress Γ_i on X_i by any method you like.
  - Regress on a few chosen covariates → **group average treatment effects** (GATEs) with valid standard errors.
  - Regress with a forest → a nonparametric CATE surface.
  - Regress on a constant → back to the ATE.
- This one construction unifies the whole section.

Causal Forests ★

Wager and Athey (2018), Athey, Tibshirani, and Wager (2019): a random forest whose splits are chosen to maximize *heterogeneity in the treatment effect* rather than in the outcome.

- **Honesty**: within each tree, one subsample chooses the splits and a *disjoint* subsample estimates the effects in the resulting leaves. This is what delivers asymptotic normality and valid pointwise intervals , without it, leaves are selected *because* they show large effects.
- **Local centering**: residualize Y and D on X first (i.e. orthogonalize), exactly as in DML.
- Output: an estimate $\hat{\tau}(x)$ for every unit, with standard errors, plus variable-importance measures for *effect* heterogeneity.
- `grf::causal_forest()`.

Reporting Heterogeneity Honestly ★

An estimated $\hat{\tau}(x)$ that varies is not by itself evidence of heterogeneity, noise varies too.

- **Calibration test**: regress the doubly robust scores on $\hat{\tau}(X_i)$ (out of fold). A slope near 1 indicates real, well-calibrated heterogeneity; a slope near 0 says your CATE surface is noise. `grf::test_calibration()`.
- **Sorted group ATEs**: rank units by $\hat{\tau}$, form quantile groups, and report the doubly robust ATE *within each group* using held-out data. Then test whether the top and bottom groups differ.
- **Pre-specify** the moderators you care about wherever possible, and label exploratory heterogeneity as exploratory.
- Effect heterogeneity is where causal ML is most useful and most abused. The honest-splitting discipline is not optional.

Where This Leaves the Course

- **One idea, three appearances.** FWL (Week 1) → double robustness and Neyman orthogonality (Week 6) → DML and causal forests (today). Each generalizes the last; none replaces identification.
- **ML relaxes functional-form assumptions, not identifying assumptions.** That is a real and substantial gain, many published identification arguments lean on linearity they never defended, but it is a bounded one.
- **The binding constraint is still the design.** A causal forest on a confounded comparison estimates heterogeneity in a bias.

Next week

Synthesis, and a review for the final exam.

References

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